



Building Multimodal Chart Accessibility for Blind and Low-Vision Individuals in the Prompt-Driven Era

As AI makes it easier than ever to generate charts, blind and low-vision individuals risk being left further behind. Multimodal approaches can offer a path toward born-accessible visualization, ensuring that prompt-driven charts can be explored and validated by senses beyond just sight.

By Pramod Chundury

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A sighted analyst types “show me quarterly revenue growth” into an AI assistant; seconds later, a line chart pops onto the dashboard. For blind or low-vision (BLV) colleagues, the same success of automation is an instant barrier: The bitmap image has no alt-text, is difficult to convert into tactile graphics [1], or has no hooks for sonification tools [2–3]. Ironically, the very tools that promise to democratize data risk widening the gulf between those who can see the pixels and those who must hear or feel the charts.

A recent survey of 28 BLV professionals exposes just how often such gaps occur [4]. BLV participants, most of them highly educated programmers or analysts, encountered accessibility problems at every stage of their workflow: importing CSVs, cleaning data, inspecting charts, and even presenting results. Heavy reliance on sighted collaborators was the norm, and several respondents said this reliance compromised their professional independence and impeded collaboration. The authors of the study conclude that the

research and developer communities must pivot from retrofitting access to creating “born-accessible” tools in which alternative representations are treated as first-class citizens [5].

The critical question, then, is how to ensure that the breathtaking speed of prompt-driven visualization does not simply automate inaccessibility at scale. Generating accessible metadata that feeds into assistive technology—devices that typically translate information to the senses of sound and touch—could be one way forward.

WHY MULTIMODALITY MATTERS MORE THAN EVER

Human vision excels because it offers bandwidth—tens of millions of photoreceptors updating in parallel. Yet decades of sensory-substitution research demonstrate that combinations of other senses can approach that informational throughput [6]. Designing for multimodal representations is crucial to reducing accessibility barriers in the future. Three channels have proven especially powerful for accessible data visualization work.



Touch captures spatial relationships. Multi-line refreshable Braille tablets, such as the Monarch, embed 10 rows of 32 pins in an area roughly the size of an iPad. Users can pan, zoom, and even toggle chart legends entirely by touch [7].

Sound reveals trend and density. Systems such as TactualPlot [2] and ChartA11y [8] map a fingertip-sized “sampling lens” to sound (e.g., pitch): The denser the data cluster under the finger, the higher the tone. Discrete taps trigger spoken summaries.

Speech provides labels, hierarchy, and data accuracy. Screen-reader systems, such as Olli, transform a chart into nested items: title at the top level, axes next, data series after that, individual data points last [9].

Together, these channels can share the cognitive load. In TactualPlot’s formative design sessions, participants reported forming a mental model from touch-based exploration (navigation) of the touchscreen and used their auditory buffer for both navigation and data tasks [2]. Tactile graphics preserve axes, tick marks, and legends—even in low-resolution grayscale—and are far faster to interpret because the familiar grid-based scaffolding links the non-visual view to the canonical visual one.

Importantly, multimodality benefits sighted users too. Haptic lane-departure rumbles help drivers keep their eyes on the road; acoustic cues in trading dashboards let analysts catch anomalies without glancing away; and video gamers rely on vibration and spatialized audio for immersion. More specific to visualization, recent research is exploring a combination of audio and visual analytics to help encode more data into interactive sonified visualization [10].

GENERATIVE AI: A DOUBLE-EDGED SWORD FOR ACCESSIBILITY

Large-language-model (LLM) pipelines can now co-author chart specifications, choose color scales, and even draft narrative captions. In the research prototype Umwelt, authors can edit a chart, its textual description, and a linked sonification track because all three derive from a single abstract data representation model [11]. The same

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architecture lets Umwelt warn users if a filter breaks the link between, say, a bar’s value and its spoken label. Such semi-automated authoring tools help blind and sighted individuals create accessible charts.

Meanwhile the LLM-powered ChartGPT shows that a single sentence (“What kind of movies are the most popular?”) can yield an executable Vega-Lite specification in seconds [12]. Such systems prove that if an AI can assemble an SVG it can at least draft a first-pass alt-text or suggest a sonification mapping. The born-accessible principle demands that accessibility metadata emerge from the same generation step as the visual marks. Doing so not only saves re-work but also keeps alternative representations semantically aligned. If a value updates, every channel updates too.

PRINCIPLES DISTILLED FROM MULTIMODAL CHART ACCESSIBILITY RESEARCH

Several converging research threads—interviews, participatory design, systematic reviews—provide guidelines for designers and tool builders to make charts accessible using multimodal approaches.

Match interaction to a fingertip. In systems such as ChartA11y and TactualPlot, a circular sampling region sized to the average index finger is used to interact with sonified charts on touchscreen devices such as smartphones and tablets. This interaction paradigm of touch-based exploration provides richer multimodal experiences (sound and speech) for perceiving charts.

Standardize auditory mappings. Us-

ers learn relative pitch scales across tasks; arbitrary remapping between charts forces relearning and increases cognitive load [2, 8].

Pair overview with drill-down. Continuous sonification conveys global trends; discrete taps or keyboard shortcuts can deliver precise values on demand. Systems such as Olli describe the chart semantics and data for screen readers by generating a hierarchical textual description of charts—think richer alt-text and a more interactive experience for screen reader users.

Preserve visual scaffolding. When considering tactile representations of visual charts, the visual scaffolds, such as including axes, tick marks, and legends, even at low tactile resolution, anchor users’ spatial reasoning.

Lean on declarative grammars. Libraries such as Vega-Lite separate “what” from “how,” making it simpler to attach non-visual encodings. LLMs are able to generate web charts with Vega-Lite and other visualization libraries’ specifications (D3.js, Highcharts, Google charts, etc.). Recent work on Tactile Vega-Lite introduces smart defaults (e.g., avoiding filled areas that merge under pin thresholding) and registers tactile channels alongside color or size [13].

These five principles resonate with the Web Content Accessibility Guidelines’ rubric—perceivable, operable, understandable, and robust—and extend it into genuinely multimodal territory. However, the onus is on the chart authors who use LLM-powered agents to embed accessibility metadata into their visualization specifications, or, at the very least, generate charts with detailed alt-text and captions.

LOOKING AHEAD: CHARTS THAT SPEAK, SING, AND CAN BE TOUCHED

What would it take for the next generation of charting assistants, whether LLM- or menu-driven, to embody those principles by default?

Accessibility artifacts can be treated as part of the codebase. Alt-text files, sonification mappings, and tactile-ready SVG layers should live beside JavaScript and CSS in version control. The Chartability auditing suite already automates color-contrast and keyboard checks [14]. A multimodal exten-

sion could do the same for sonification channel conflicts or missing Braille labels.

Leverage semi-automatic authoring. Designers of Umwelt report that users are comfortable letting the system draft a first-pass description while retaining editorial control for encoding, technical vocabulary, and cultural nuance [11]. Automating chart creation using LLM-powered tools can feel like a superpower, but this should not be at the cost of accessibility issues flowing through the data analysis workflow.

Build on open grammars. Declarative visualization specifications make the chart creation process simpler for data analysts. The designers of such libraries need to think about multimodal assistive technology as output and think beyond only the visual sense. As evidenced by the recent Tactile Vega-Lite work, visualization grammars can nudge authors to fill in the accessibility data while charts are being programmed. In one of our recent studies, we needed tactile-friendly graphics, exporting grayscale Vega-Lite charts as PDFs turned out to be faster than redrawing them manually because the grammar already contained the semantic structure required.

Educate early. BLV students often meet their first histogram or scatter plot in a classroom that cannot render it non-visually. BLV students work with teachers of students with visual impairments who often remediate course materials to make them accessible. Standard classroom materials and environments from elementary to college levels are often not accessible. Born-accessible teaching modules, coupled with tactile or audio scaffolding, can begin to close that gap, but mainstream analytics curriculums still assume eyesight. Exposure to multimodal tools early in education can seed skills that carry into the workforce. Sonification, supported by off-the-shelf hardware, can be effective for certain data visualization tasks. However, it is still in the nascent stage, as BLV individuals have not encountered interactive sonification systems as part of their training and education.

Research still faces open questions. How can auditory scales adapt automatically when a user switches from

one chart type or dataset to the next? What haptic vocabulary will be understandable for visualization tasks? How will LLMs avoid hallucinating alt-text for charts? Early answers are emerging from cross-disciplinary collaborations that blend HCI, audiology, tactile engineering, and machine learning. Projects inspired by the sensory-substitution model outlined in Tactual-Plot foresee cross-modal “zooming,” where a pinch gesture on a tactile tablet tightens the pitch range to expose fine differences without losing the audio contour.

CONCLUSION

Generative AI now drafts charts at incredible speeds. Our responsibility is to ensure that accessibility travels at the speed of creation. The hardware is here already: tactile-ready displays, web-native spatial audio, declarative grammars that treat visual marks and chart semantics as first-class data, and LLMs that can narrate what they draw. The task ahead is to meld those pieces into default behavior so that authors scarcely notice the extra lines of metadata generated alongside their charts. Imagine the next round of office analytics: A prompt yields not just a line chart but a bundle of metadata for alternate representations. The sighted project manager skims the picture; a blind colleague taps her Monarch and feels or hears the trend. No one waits, no one translates, and better collaboration ensues. Vision is a fleeting sense, and any impact we can have for BLV individuals can help all. The important question is not whether we can build multimodal tools, but whether we will prioritize them—ensuring data trust and equity as AI tools accelerate the field. It begins with a simple shift in mindset: A chart is not finished when the pixels look good, but when every sense that needs to can understand the data story being told.

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Biography

Pramod Chundury is a recent Ph.D. graduate from the College of Information Studies (iSchool), and a student member of the Human-Computer Interaction Lab at the University of Maryland, College Park. His research explores ways to provide equitable access to information and data and consequently help individuals make better data-driven decisions. He studies multimodal experiences with data visualization, such as using visualization, sonification, physicalization, and conversational experiences to assist human cognition. He is also interested in improving collaboration between individuals with and without disabilities and employs mixed-methods to design and build assistive technology for data analysis.

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